

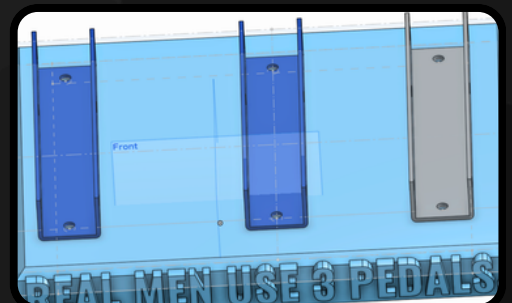
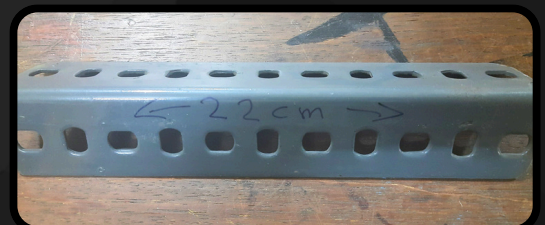
FFB RACING SIM - BUILD GUIDE:

NOTE:

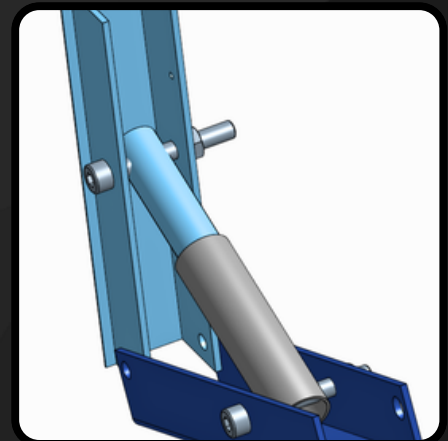
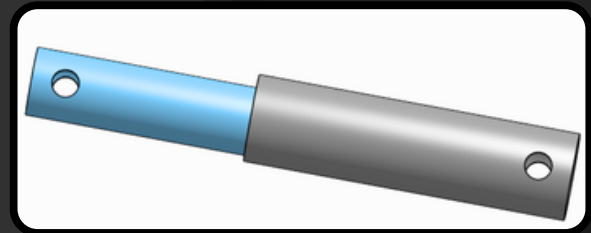
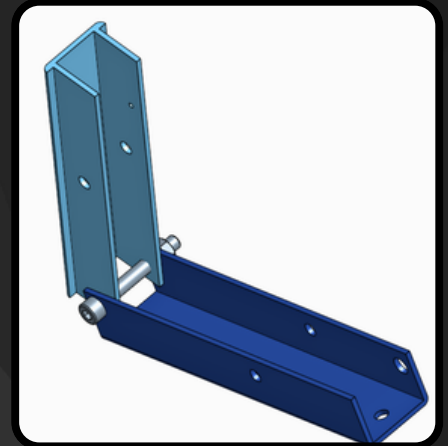
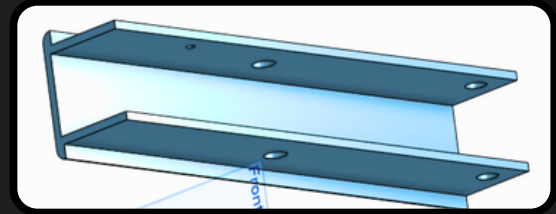
- This is just a guide, you are free to change the dimensions or components if you like, provided you know what you are doing.
- Every step does not need to be exact, a close enough replication works, as lot of the stuff I used were sourced locally:)

PEDALS:

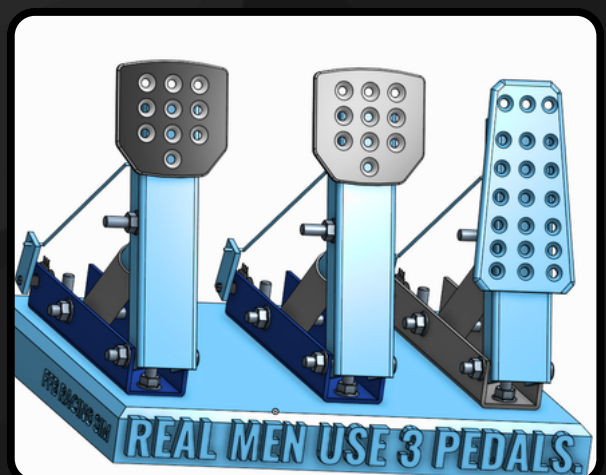
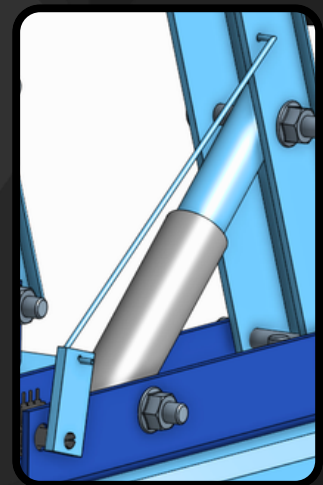
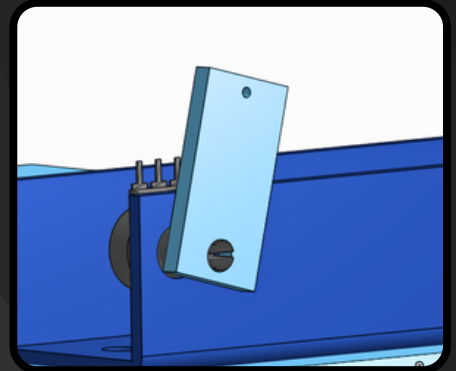
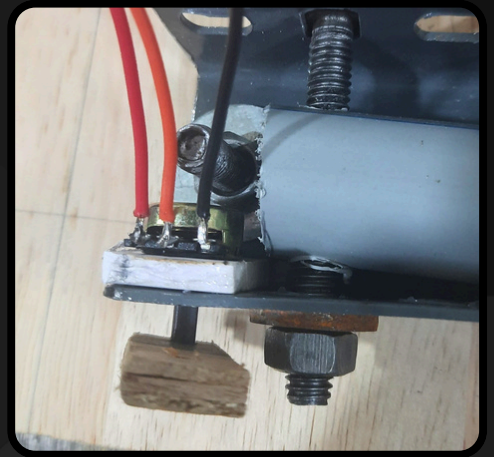
- Get your wooden base of dimension 36cm by 30cm (if you are anchoring it somewhere), or 36cm by 45cm if you are not anchoring it.
- Add 4 beams of about 3cm width to each edge.
- With the help of 22cm L-Bracket, mark 3 sets of holes on the base for each pedal. (10cm b/w the brake and gas pedal ; 17cm b/w the brake and the clutch).
- Now drill the holes with an 8mm drill bit.
- Take 2 of the 22cm L-Brackets and overlap them, to form a U-Shape, now mount them on the base with m8 nuts and bolts.
- Repeat the previous step thrice.



- Take 2 of the 19cm L-Brackets and overlap them to form a U-shape and fasten them together with m8 nuts and bolts.
- Now join the upper pedal to the base with m10 nuts and bolts.
- You can use a pair of pliers to bend the base L-Bracket outwards if needed.
- Take out your $\frac{3}{4}$ inch and 1 inch pvc pipes and cut them to 10cm each(3pcs for each dia), this length depends on your spring, I am using a 4inch one. It also depends on how much pressure you want to apply on the pedals, so definitely feel free to customize.
- Now drill out one hole through each pipe with an 8mm drill bit.
- Now take the 1inch pipes and mount them to the base L-Bracket with m8 nuts and bolts, and do the same for the $\frac{3}{4}$ inch pipes, but to the upper pedal.
- Repeat this step thrice.
- Now use epoxy to make a seal on the other end of each $\frac{3}{4}$ inch pipe.
- Once that dries, take out your spring and put it in the 1inch pipe.
- Now use the $\frac{3}{4}$ inch pipe to enclose the spring, and use zipties or something to limit the pedals movement so the springs don't come out.

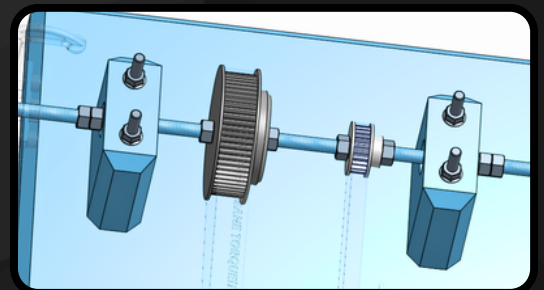
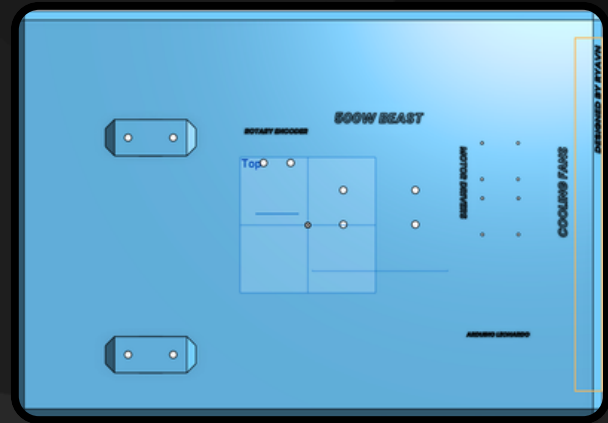


- Now press on each pedal to ensure they work smoothly.
- Take out the potentiometer and mount it to the end on the base L-Bracket, if the cutouts on the bracket is too big, use foam board or wood to fill in that space.
- Now cut out 3 pieces of wood with holes drilled on either end for the horns and attach it to the potentiometer shaft.
- Repeat this thrice.
- Now take your push rod and cut it accordingly while keeping in mind to leave some extra at each end as it will have to be bent.
- Repeat this thrice.
- Use the push rod and put one end in the potentiometer horn while the other is attached to the upper pedal.
- Repeat this thrice.
- Now slap on the pedal plates for each, this is upto you as it doesn't affect functionality.
- Now solder wires onto the potentiometer, following the schematics.
- We are done with the pedals!!!

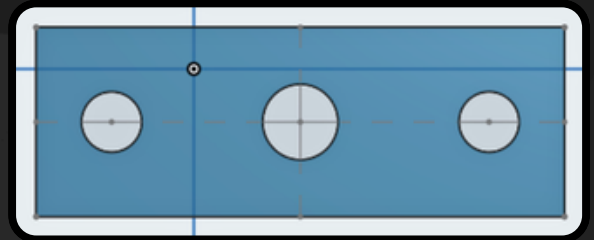
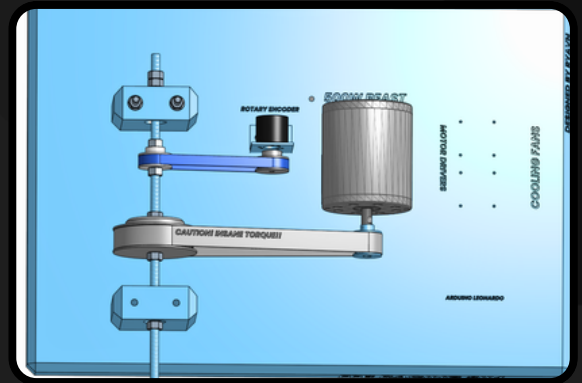


WHEEL-BASE:

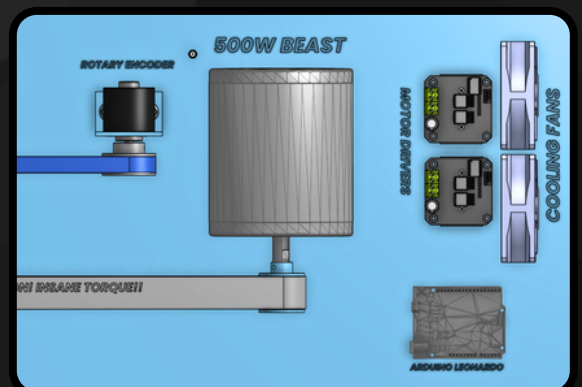
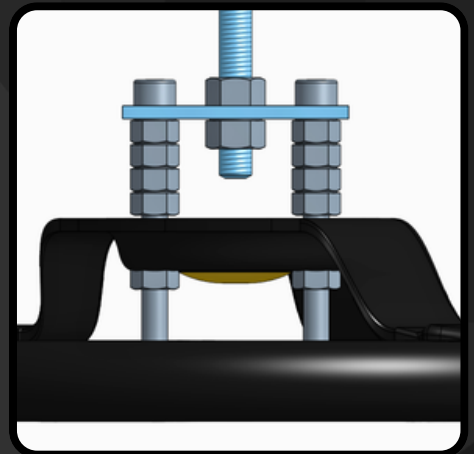
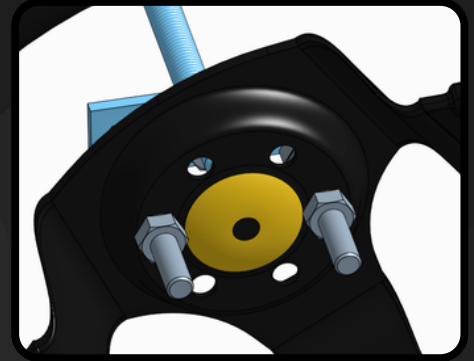
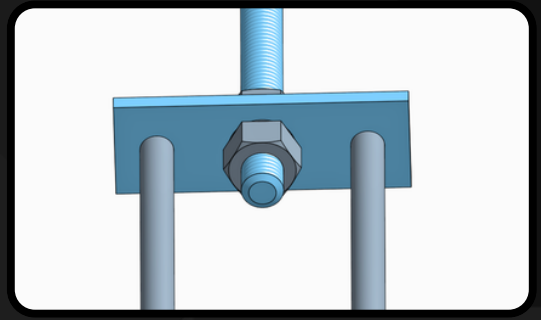
- Take a wooden base of dimension 40cm*65cm
- Take the wooden steering shaft mount of dimensions 9*10cm with the bearing mounted in it.
- Drill 2 holes in the wooden mount through out with an 8mm drill bit.
- Now with the help of the steering shaft mount, mark 2 holes about 7 to 8cm away from the long edge (please refer to the onshape link or cad file for a better understanding).
- Now with the help of the same mount, mark 2 more holes about 24 to 25cm away from the previous holes.
- Now drill these holes with an 8mm drill bit.
- Mount one steering shaft mount to the holes closer to the edge with m8 nuts and bolts.
- Now insert the threaded rod into the bearing and position the 2nd steering shaft mount on the other set of holes, but don't mount it yet.
- Now Take out the 60 teeth and 22 teeth pulleys and fasten them to the threaded rod as shown in the CAD. Ensure they are properly secured with m10 bolts.



- Now insert both the pulleys' belts onto the rod.
- Mount the other steering shaft mount with m8 nuts and bolts.
- Take out your motor and encoder (with mount).
- Mount the 15teeth pulley onto the motor shaft.
- Use the longer timing belt by positioning it on the 60 and 20 teeth pulleys and mark 4 holes for mounting the motor (ensuring the belt is tight and fits snug on the pulleys), ensure these holes are offset to the direction away from the threaded rod by about 0.5cm to induce tension in the belt.
- Do the same with the rotary encoder and mark 2 holes.
- Now drill out all these holes with an 8mm drill bit.
- Mount the rotary encoder and motor to their mount holes accordingly with m8 nuts and bolts.
- If the belt doesn't have enough tension, or is too tight (please refer to yt tutorial on checking belt tension), shift the corresponding pulley a bit on the threaded rod, and drill the mount holes again, but a bit closer to the shaft.
- Now make the steering mount, I used a rectangular metal piece and drilled 3 holes: one 10mm one in the center and



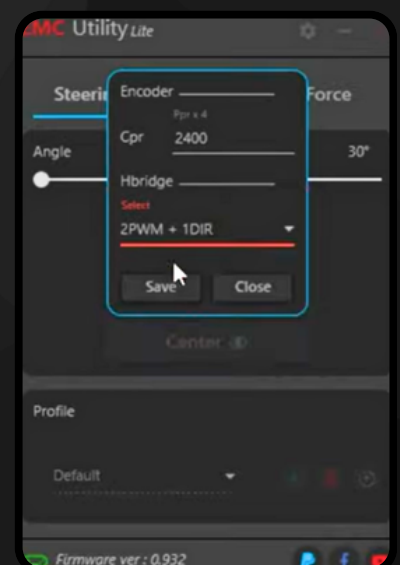
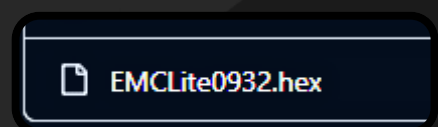
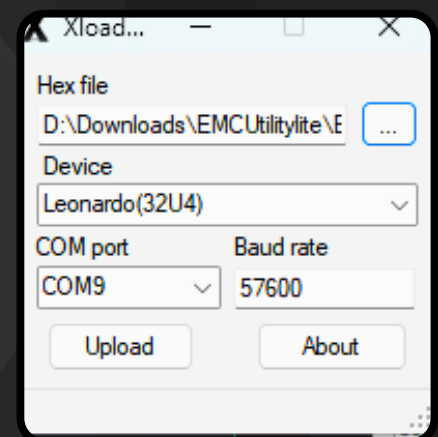
- Now make the steering mount, I used a rectangular metal piece and drilled 3 holes: one 10mm one in the center and two 8mm ones equidistant from the center on both the sides (This depends on your steering wheel).
- Now push the center hole onto the steering shaft (threaded rod) and use m10 nuts on either side to fasten it.
- Insert two m8 bolts and use a few nuts to act as spacers.
- Now mount your steering wheel onto these two m8 bolts and fasten it with m8 nuts.
- Give the wheel a spin to check if it moves smoothly:)
- Now take both of your motor drivers and position it near the motor.
- Now mark out the mount holes for the motor driver, and drill them out with a 4mm drill bit
- Mount them with m4 nuts and bolts.
- Now take the cooling fans and position it near the heatsinks of the motor driver, you can now glue them in place or use L-Brackets to mount them accordingly.
- Now take your Arduino Leonardo and position it a bit below the motor driver and mark out 4 holes.
- Drill the holes with a 4mm drill bit and mount in with a set of m4 nuts



- Drill the holes with a 4mm drill bit and mount it with m4 nuts and bolts. You can also use glue!
- The position of the Arduino, Motor driver and the cooling fans aren't that crucial, as long as the wiring is neat and they have proper ventilation:)
- Now wire them up according to the schematics.
- You have officially finished the Wheel Base!!!

CONFIGURING:

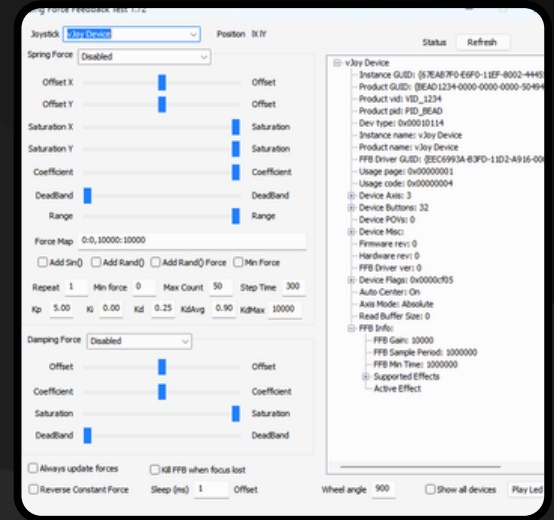
- Download all the required files from my GitHub Repo.
- Connect your Arduino to your PC, and open Xloader.
- Select the downloaded hex file from my repo, leonardo as the device.
- Now short the reset and GND pins briefly on the leonardo and remove, this will show a new COM port, which you have to select.
- Now upload the firmware.
- Now open settings, and set the cpr value to your ppr value*4, mine is 1000 so $cpr = 4000$ and save it.
- Now go to the pedals tab for calibration: Keep a pedal pressed down entirely, and click max; now let go of it and click min.
- Repeat this thrice.



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- Now go to the pedals tab for calibration: Keep a pedal pressed down entirely, and click max; now let go of it and click min.
- Repeat this thrice.
- Now go to the main page, now center your steering wheel physically and click center wheel.
- You can set the max degree of rotation for the steering wheel depending on your game/sim.
- You can also go to the force tab and play around with the settings, till you find it comfortable.

TESTING:

- You must always keep the EMC app open when running the wheel.
- Select EMC for joystick.
- You can use the Wheel Check software to test if the wheel is running smoothly.
- You can open any compatible game/sim and connect the controller like any other wheel:).
- Calibrate all the axes and buttons and you're good to go!!
- You've officially completed the build!!!!



-BY Ryaan Ahmed:)